

7	(a)	$P = \frac{Nhf}{t}$ $3.8 \times 10^{-3} = (N/t)(4.7 \times 1.6 \times 10^{-19})$ $N/t = 5.1 \times 10^{15} \text{ s}^{-1}$	C1 A1
	(b)	$I = \frac{Ne}{t}$ $0.800 \times 10^{-8} = (N/t)(1.6 \times 10^{-19})$ $N/t = 5.0 \times 10^{10} \text{ s}^{-1}$	A1
	(c)	- Most of the photons are reflected from the metal surface. - Atoms are made up of mostly empty space and thus the probability of a photon hitting a surface electron is very small. Instead, photons may hit electrons which are deeper below the surface, and these electrons may lose all its kinetic energy on its way up to the	Any two

		surface. (due to collisions with the metallic ion core) 3. Electrons emitted may travel at an angle away from the normal and may not be collected at the anode.	
	(d)	$eV_s = \frac{1}{2}mv^2$ $(1.6 \times 10^{-19})(0.500) = (1/2)(9.11 \times 10^{-31})v^2$ $v = 4.19 \times 10^5 \text{ m s}^{-1}$	C1 C1 A1